Who Pays When the Wholesale Price Moves?

Contracting, Gap Charges and Pass-Through in World Electricity Markets

Kilambi Research

Kilambi Research Working Paper No. KR-WP-2026-37
November 2026

Abstract. When a wholesale electricity price falls, generators are often assumed to lose and customers to gain. In most of the world's electricity systems that assumption fails, because most electricity is not sold at the wholesale price. It is sold under private contracts, under publicly backed contracts that settle the difference between a contract price and the market price, or under regulated tariffs that recover costs with a lag. This paper maps, market by market for the largest electricity systems in the world, how generators are paid, who absorbs the difference when market prices move, and how much of a price change reaches household and large-user bills. A computed worked example traces a $20/MWh price decline and increase through a stylized market: the average consumer cost moves only $6.00/MWh in either direction in year one, while the gap levy that funds contracted generation rises $5.00/MWh in the decline and turns negative in the increase. Market sections cover China, the United States by market design, the European Union through five member states and the 2024 rule mandating two-way contracts for difference, India, Japan, Brazil, South Korea, Canada with Ontario's Global Adjustment as the reference case, the United Kingdom, Australia, the Nordic market, South Africa, Mexico and the Philippines. Prevalence is estimated from covered volumes rather than whole-country classification. Against 2024 world electricity consumption of 30,856 TWh, regulated tariffs settle 34.8 percent, private contracts 21.1 percent, spot and merchant sales 9.0 percent, and publicly backed contracts 2.1 percent; the publicly backed volume is also exposed to spot settlement (an overlap of 2.1 percent of world consumption), and 33.1 percent of world consumption cannot be classified from published covered volumes.

Keywords: electricity markets; contracts for difference; feed-in tariffs; Global Adjustment; cost pass-through; regulated tariffs; tariff deficit

JEL codes: L94; L51; Q48

Non-technical summary. Electricity bills do not move with the wholesale price in the way most commentary assumes. In Ontario, when the market price of electricity falls, a charge called the Global Adjustment rises to cover what generators are owed under their contracts, so the saving on the energy itself is partly handed back. Ontario is unusual only in printing the mechanism on the bill. This paper checks how the same situation is handled in the largest electricity markets in the world. It finds that most electricity everywhere is sold under contracts or regulated prices rather than at the market price, that the difference between contract and market prices is paid by customers or taxpayers under many local names, and that well-designed versions of the charge run in both directions: when market prices spiked in 2021 and 2022, contract holders in Britain paid money back and the charge fell to zero or below. The practical consequence is that a fall in wholesale prices delivers a smaller and slower bill reduction than the headline price move suggests, and the size of the offset depends on choices each market has made about contract design, who is exempt from the charge, and how often it is reset.

Contents

1 Introduction **4**

2 Institutional background **5**

3 Related literature **6**

4 Data **6**

5 Methodology: four channels and the incidence variables **7**

5.1 The four channels 7

5.2 The incidence variables 8

5.3 Method 9

6 The worked example **9**

6.1 Setup 9

6.2 The decline: spot falls from $50 to $30 10

6.3 The increase: spot rises from $50 to $70, and the asymmetry 10

6.4 The no-spot variant: a cost shock inside a regulated utility 10

7 Market evidence **12**

7.1 China: a capped market, a frozen surcharge, and a two-way turn 12

7.2 The United States: three systems in one country 12

7.3 The European Union: five member states and a common rule 13

7.4 India: the gap inside discom tariffs, deferred as regulatory assets 15

7.5 Japan: the surcharge that is net of the spot price 16

7.6 Brazil: regulated auctions and a free market share the gap charge 16

7.7 South Korea: a single-buyer pool and a certificates overlay 17

7.8 Canada: Ontario's Global Adjustment, and the regulated rest 18

7.9 United Kingdom: a two-way CfD with a floored levy 18

7.10 Australia: merchant energy, certificates, and a taxpayer collar 19

7.11 The Nordic market: merchant by default 20

7.12 South Africa: building the gap charge into the new market 20

7.13 Mexico: the gap absorbed by the Treasury 21

7.14 The Philippines: FIT-All and the collection holiday 21

8 Summary of mechanisms **21**

9 Prevalence: how much of world electricity each channel settles **28**

10 Discussion **29**

11 Limitations **31**

12 Conclusion **31**

A Data appendix: sources and verification status **32**

List of Tables

1 Markets examined: consumption and data sources. 23

2 Worked example: generator revenue and gap payments by scenario. Stylized market, 100 TWh, consumption held constant. USD millions per year. 24

3 Worked example: what reaches the bill. USD per MWh unless stated. 24

4 Worked example: incidence and timing by channel 24

5 Market-by-market summary: channels, schemes, payer, direction, reset lag, and incidence by customer class. 25

6 Prevalence by covered volume, 2024. TWh and percent of world electricity consumption (30,856 TWh). 28

List of Figures

1 Worked example: the gap levy and the average consumer cost. Top panel: the levy that funds the publicly backed channels rises when the spot price falls and turns negative when the spot price rises above the CfD strike, under both levy bases. Bottom panel: the average consumer cost moves $6.00/MWh against a $20.00/MWh spot move. Source: Desk computation from `data/worked_example_assumptions.json` (Modelled, C). Note: Stylized market of Section 6.1; consumption held constant at 100 TWh. . . 11

2 Japan's Renewable Energy Surcharge (yen/kWh), METI annual determinations, FY2022–FY2025. The FY2023 collapse reflects the FY2022 spot-price spike raising avoidable costs; the rebound follows as wholesale prices normalized. FY2023 rate: (Observed, B); other years (Observed, A). 17

3 Ontario: Global Adjustment per $1 of wholesale commodity value, by year, 2005–2025. The dashed line marks parity. The ratio is highest when market prices are lowest, the inverse signature of a contract-settlement charge. Source: (Reconstructed, A) from IESO settlement data. 19

4 Shares of 2024 world electricity consumption by settlement channel, covered-volume method. Whiskers show the range from varying each input between its low and high entries. Source: (Reconstructed, B). 29

1 Introduction

Wholesale electricity prices moved violently in the first half of this decade, and the movement produced a recurring public puzzle. Market prices fell, and bills did not fall by anything like the same amount. Market prices spiked in 2021 and 2022, and in a few systems bills were partly shielded while in others they rose almost in full. The gap between the price movement and the bill movement is not measurement error and not, in the main, supplier margin. It is contract structure. Most electricity in most systems is not sold at the wholesale price at all. It is sold under private contracts at fixed prices, under publicly backed contracts that pay the difference between a contract price and the market price, or under regulated tariffs that recover audited costs months or years later. The wholesale price is the marginal price of a minority of volume, and the bill is an average over all of it.

Ontario states the mechanism on the customer's bill. Generators in Ontario hold contracts or regulated rates that fix what they are paid per megawatt-hour. The market pays them the hourly spot price. The difference is settled monthly through the Global Adjustment (GA), a charge on consumption. When the spot price falls, the GA rises by construction; over the ten years to September 2026 the GA collected C$94.5B against C$47.1B of market (commodity) revenue, or $2.01 of GA for every $1 of commodity (Reconstructed, A). Ontario is the reference case for this paper because the mechanism is itemized there. The paper's claim is that the mechanism itself is close to universal, and that markets differ mainly in whether it is itemized, who pays it, how often it resets, and whether it runs in both directions.

The paper answers three questions for each major market. First, how is electricity sold: at spot prices, under private contracts, through publicly backed contracts, or under regulated tariffs, and in what proportions where volumes are published. Second, when the wholesale price falls or rises, who gains and who absorbs the difference, and does any payment reverse sign. Third, how much of the change reaches the bills of households and of large commercial and industrial users, and when.

Three findings recur. First, the pass-through of a wholesale price change to the average bill is partial and slow in every large system examined, because the spot-priced share of volume is a minority almost everywhere; in the paper's worked example a $20/MWh spot move changes the average consumer cost by $6.00/MWh in year one. Second, where the gap between contract and market prices is settled through a named charge, the charge is countercyclical by construction: it rises when prices fall and falls, in two-way designs to zero or below zero, when prices rise. Third, suppressing the charge does not remove the cost. Spain, Portugal, India and China each show the same device in different dress: when the charge is held below the gap, the difference accumulates as a tariff deficit, a regulatory asset or a fund deficit, and is collected from future customers, with financing costs, later.

The contribution relative to the existing literature is accounting rather than theory. The components are well studied separately: contract design and missing money, auctioned support schemes, cost pass-through in regulated industries, and national case studies of tariff deficits. What has not been assembled in one place is a market-by-market incidence account on a common template, with a prevalence estimate built from covered volumes instead of whole-country labels. Section 9 replaces the whole-country classification used in a scoping review prepared for this project, which assigned each country to a single class and produced a 19/72/8 split of world consumption; the covered-volume method reports overlap and an explicit unknown share, and the two estimates are reconciled there.

The rest of the paper proceeds as follows. Section 2 describes how electricity is sold and settled, for a reader who does not follow the industry. Section 3 reviews the related literature. Section 4 documents the data. Section 5 sets out the framework: four sales channels and the variables that determine incidence. Section 6 computes the worked example. Section 7 reports the market-by-market evidence, ordered by electricity consumption, and Section 8 collects it in a standardized table. Section

9 estimates prevalence from covered volumes. Section 10 discusses what the mechanics imply, Section 11 states the limitations and what evidence would change the conclusions, and Section 12 concludes. Appendix A is the data appendix, with the source and verification status of every load-bearing figure.

2 Institutional background

Electricity is sold in two physical layers that are easy to confuse. The first layer is dispatch: a system operator instructs generators to produce, minute by minute, so that supply equals demand on a network that stores almost nothing. The second layer is financial: contracts and tariffs that decide who pays what for the energy dispatched. The wholesale spot price belongs to the first layer. It is formed in day-ahead and real-time markets, or by a dispatch algorithm in systems without an explicit exchange, and it prices the marginal megawatt-hour. Bills belong to the second layer, and the second layer is where most money is settled.

Four features of the financial layer matter for what follows.

First, a large share of generation never faces the spot price. A generator with a fixed-price contract receives the contract price; the spot price determines only the settlement between its buyer and the market. A generator in a vertically integrated utility receives a regulated revenue requirement, computed from audited costs plus an allowed return, and there may be no meaningful spot price in its system at all. Spot exposure is the exception that financial reporting tends to treat as the rule, because spot prices are published daily and contract prices are not.

Second, where governments have procured generation, they have usually bought revenue certainty rather than energy. Feed-in tariffs, feed-in premiums, contracts for difference (CfDs), renewable certificate obligations and auctioned power purchase agreements all fix, floor or top up the generator's revenue per megawatt-hour. The state rarely takes delivery of the electricity. The generator sells into the market, and a settlement entity pays or collects the difference between the contract revenue and the market revenue. That difference has to be funded, and the funding route is the subject of this paper: a per-kWh levy on consumption, a component inside regulated tariffs, the general budget, or a deferral that becomes debt owed by the system.

Third, retail supply inserts a second contract layer between generators and customers. Suppliers buy energy in advance through bilateral contracts and exchange-traded hedges, and sell to households on tariffs that reprice at most a few times a year. A wholesale price move therefore reaches a household bill through three filters: the supplier's hedge book, the tariff reset calendar, and any regulated price cap. Large industrial users sit closer to the market: they buy on indexed contracts, self-supply, or negotiate directly, and they are the customers most often granted exemptions from policy levies, which shifts the levies onto everyone else.

Fourth, regulated systems replace the settlement problem with a timing problem. A utility that recovers costs through tariffs files evidence, the regulator sets allowed revenue, and the difference between forecast and actual costs is trued up later, through a fuel adjustment clause where one exists or at the next rate case where one does not. The interim difference is carried by someone: the utility's shareholders, a deferral account owed by future customers, or the state. Regulated systems therefore have the same incidence question as contract systems, answered on a slower clock.

Ontario combines the first three features in itemized form. The Independent Electricity System Operator (IESO) settles the spot market. Contracted and regulated generators receive their contract or regulated amounts. The shortfall between what they are owed and what the market pays them is the Global Adjustment, charged to consumers monthly, with different settlement treatment for large

(Class A) and smaller (Class B) customers. Section 7.8 uses Ontario as the reference case against which the less transparent versions of the same arrangement are compared.

3 Related literature

The paper sits at the intersection of four literatures, each of which studies one part of the mechanism in isolation.

The first is the market-design literature on restructuring. Joskow (2008) draws the central lesson that wholesale spot markets were layered onto systems in which most volume remained under contract or regulation, and that the performance of the spot layer cannot be read as the performance of the system. Borenstein et al. (2002) measure inefficiencies in California's restructured market, and Bushnell et al. (2008) show how vertical arrangements and contracting muted spot-market incentives in restructured US markets. The contracting share, not the spot price, is the object that decides who bears price risk; this paper takes that point as its accounting starting point and measures the shares.

The second literature studies support instruments for low-carbon generation. Menanteau et al. (2003) compare price-based and quantity-based promotion and identify the risk allocation that distinguishes them: a feed-in tariff places price risk on the buyer of last resort, a certificate scheme leaves it with the generator. Fabra (2021) reviews auctioned contracts and argues that two-way contracts for difference have become the default allocation of price risk to the public balance sheet or the bill. Newbery (2018) treats the missing-money problem that capacity and contract mechanisms are built to solve. What these papers evaluate as design, this paper follows as incidence: the same contract that removes price risk from the generator places it, arithmetically, on levy payers, tariff customers or taxpayers.

The third literature estimates cost pass-through. Fabra and Reguant (2014) estimate the pass-through of emissions allowance costs into Spanish electricity prices, and Sijm et al. (2006) review pass-through rates for the EU ETS. Pass-through in that literature is a property of market competition. In this paper the relevant pass-through is mostly administrative rather than competitive: levy resets, tariff true-ups and regulated price caps transmit costs on published calendars, and the empirical question is the size and lag of the transmission, not its sign.

The fourth literature is national and institutional. Espinosa (2013) documents the Spanish tariff deficit, the case in which a suppressed gap charge accumulated as system debt. Case material of the same kind exists for the Ontario Global Adjustment in Ontario Energy Board and IESO publications, for the German EEG in Bundesnetzagentur monitoring reports, and for the British CfD regime in Low Carbon Contracts Company reporting; these primary sources are used as data in Section 7 and catalogued in Appendix A rather than reviewed here.

Two gaps motivate the paper. The design literature rarely follows a contract through to the household and industrial bills that fund it, and the case literature is organized country by country, so the same settlement device appears under unrelated names and is not counted twice or compared. A common template and a covered-volume count are the response.

4 Data

The evidence is institutional and published, not sampled. Three source families are used.

Consumption and volumes. The denominator and the market sizes are electricity demand for 2024 from Ember's Global Electricity Review, the latest full year with a complete country set at the time of writing. Covered volumes by channel (the terawatt-hours actually settled through each

arrangement) come from the publications of regulators, market operators and scheme administrators: the Low Carbon Contracts Company for British CfD generation, METI and OCCTO for Japanese FIT/FIP purchases, the China Electricity Council and the National Energy Administration for China's market-traded volume, CCEE and ANEEL for Brazil's regulated and free contracting environments, the Clean Energy Regulator for Australia's Renewable Energy Target, ERCOT and the US Energy Information Administration for the United States split, and the IESO for Ontario. Each input is recorded in `data/prevalence_inputs.csv` with its year, source and verification status; Appendix A reproduces the catalogue.

Mechanics. Payment determination, recovery routes, reset calendars and exemption rules are taken from the scheme documents themselves: statutes, regulator decisions, published tariff orders and scheme administrator guidance. Where a mechanic could only be established from secondary reporting, the text says so and Appendix A grades the figure accordingly.

Ontario. The Ontario reference figures are computed from the estate's parsed IESO tables (generation capability, real-time prices and the Global Adjustment by component), not from published summaries: over October 2016 to September 2026 the GA totalled C$94.5B and market commodity revenue C$47.1B (Reconstructed, A). The volume-weighted ratio, $2.01 of GA per $1 of commodity, is the figure used in Section 7.8. The simple average of annual ratios over a comparable span is materially higher (about $3.50 for calendar 2016 to 2025), because it overweights years such as 2020 in which commodity revenue was small; the paper reports the volume-weighted figure throughout and flags the distinction once, here.

Definitions follow the sources. "Consumption" is electricity demand as Ember reports it (generation plus net imports, net of losses where the source nets them); covered volumes are generation or settlement volumes as the publishing body defines them, and the mismatch between a generation-side covered volume and a consumption-side denominator is one component of the unknown share in Section 9. Currencies are reported as published, with conversions only where a comparison requires them and stated at the point of use. Table 1 summarizes the markets examined.

5 Methodology: four channels and the incidence variables

5.1 The four channels

Every megawatt-hour generated is settled through one of four channels. The channels are defined by how the generator's payment is determined, not by technology and not by ownership.

Channel S: spot or merchant. The generator sells at the wholesale market price (day-ahead, real-time, or a dispatch-based price in systems without an exchange) and keeps what the market pays. Payment is determined hour by hour. The generator bears price risk in full, and so, through their suppliers, do customers on indexed retail contracts.

Channel P: private contracts. The generator sells under a bilateral contract (a corporate or utility power purchase agreement, or an exchange-traded hedge) at a price agreed between the parties. The generator's revenue is the contract price; the market price determines the profit or loss of the counterparty, typically a supplier that resells to final customers. Price risk is transferred, not removed: it sits with the counterparty and is priced into the retail contracts the counterparty writes.

Channel G: publicly backed contracts. A public body (a state counterparty, a regulator, a system operator acting under statute) guarantees the generator a revenue per megawatt-hour: a feed-in tariff, a feed-in premium, a contract for difference, a certificate obligation whose price tops up market revenue, or a state-procured PPA. The generator sells its energy into the market where

a market exists. The difference between the guaranteed revenue and the market revenue is settled by the scheme and funded through a levy on consumption, a component of regulated tariffs, the state budget, or a deferral. The local names differ (Global Adjustment, EEG settlement, Supplier Obligation, FIT surcharge, CDE, FIT-All); the settlement is the same object.

Channel R: regulated tariffs. There is no separate contract price. A utility that owns generation, or buys under cost-plus arrangements approved by its regulator, recovers its revenue requirement (audited costs plus an allowed return) through tariffs. Where a wholesale market exists it prices, at most, the utility's external purchases and sales. The gap concept does not apply directly; the corresponding question is how a change in the utility's generation or procurement costs travels through fuel adjustment clauses and rate cases to tariffs.

A megawatt-hour of consumption is assigned to the channel of the generation that serves it, on a pro-rata basis within each market, following published volume data where they exist (Section 9). Channels coexist in almost every market: the United States contains all four, and so does Germany.

5.2 The incidence variables

Within a channel, five variables determine what a price change does to bills.

1. **Direction.** A two-way settlement (a CfD, the Ontario GA in aggregate) pays the generator when the market price is below the contract price and collects from the generator when it is above. A one-way settlement (a floored premium, a fixed feed-in tariff, a certificate top-up) pays when prices fall and stops paying when prices rise; the generator keeps market revenue above the guarantee. One-way designs therefore truncate the customer's gain from a price spike, two-way designs transmit it.

2. **Recovery route.** The identical settlement is a different burden depending on whether it is collected per kilowatt-hour on all consumption, per kilowatt-hour with exemptions, inside an unitemized tariff, or from general taxation. Levy funding is proportional to consumption and regressive relative to income; budget funding follows the tax structure; tariff funding follows the rate design (fixed versus volumetric charges, class allocations).

3. **Reset frequency and lag.** Levies reset on published calendars: monthly in Ontario, annually for the Japanese FIT surcharge, quarterly for the British CfD Interim Levy Rate. Regulated tariffs reset at rate cases, with fuel adjustment clauses shortening the lag for fuel and purchased-power costs in some jurisdictions. The lag decides who finances the interim difference.

4. **Customer-class differentiation.** Energy-intensive users receive reduced levies or exemptions in Germany, France, Poland and elsewhere, and Ontario settles large customers (Class A) on a different basis from smaller customers (Class B). Every exemption raises the rate paid by the non-exempt base. Indexed retail contracts, more common for large users, transmit market prices quickly; regulated default tariffs, more common for households, transmit them slowly or not at all within the tariff period.

5. **Deferral.** Where the settlement or tariff is held below the amount due, the difference accumulates as a liability of the system (a tariff deficit, a regulatory asset, a fund deficit) owed by future customers. Deferral changes the timing and the financing cost of the charge, not its existence.

5.3 Method

The market evidence in Section 7 follows a single template per market: sales channels and covered volumes; payment determination; recovery route; direction; reset timing; incidence by customer class; and the traced effect of a price fall and a price rise. Mechanics were verified against regulator, market operator or scheme administrator publications wherever those could be read; figures that rest on secondary sources only are flagged in the text and catalogued in Appendix A.

The worked example in Section 6 is a computation, not a calibration. A stylized market of 100 TWh with stated channel shares and contract prices is shocked twice, a \$20/MWh decline and a \$20/MWh increase from a \$50/MWh baseline, with consumption held constant. All figures are computed from a single assumptions file (`data/worked_example_assumptions.json`) by a saved script, and the accounts are closed by construction: in every scenario, consumer outlay equals generator revenue. A second computation traces a cost shock inside a regulated utility, the no-spot variant the brief requires.

The prevalence estimate in Section 9 sums published covered volumes by channel. Its inputs are saved as an auditable table (`data/prevalence_inputs.csv`) with, for each entry, the market, the channel, the covered volume in TWh, the reference year, the source and the verification status. Consumption that published sources do not classify is reported as unknown rather than allocated by assumption. Where the same volume is counted under two channels (a publicly backed contract inside a restructured market whose energy is also sold at spot), the overlap is reported separately and shares are computed on de-overlapped volumes.

This is a retrospective analysis; there was no pre-registered plan. The conclusions would be falsified by evidence that, in the markets examined, gap settlements do not move against wholesale prices (for example, levy series that are uncorrelated with the sign of the contract-minus-market difference), or that covered volumes under Channels G and R are small relative to spot volumes in the largest markets.

6 The worked example

6.1 Setup

The stylized market consumes 100 TWh a year, held constant in every scenario. Generation is sold through the four channels of Section 5, with a fifth row separating the two publicly backed designs because their behaviour differs:

- Merchant (Channel S): 30 TWh sold at the spot price.
- Private PPA (Channel P): 20 TWh at a fixed \$55/MWh.
- Two-way CfD (Channel G): 15 TWh, strike \$60/MWh. The generator sells at the market price and the scheme settles the difference.
- One-way premium (Channel G): 10 TWh, floor \$80/MWh. The generator sells at the market price and receives a premium of $\max(80 - P, 0)$.
- Regulated cost of service (Channel R): 25 TWh at an allowed cost of \$58/MWh, recovered in tariffs.

The baseline spot price is $50/MWh. Two shocks are traced: a decline to $30/MWh and an increase to $70/MWh. All values are assumptions chosen for arithmetic transparency (ASSUMED, C); the computation, including the closure check that consumer outlay equals generator revenue in every scenario, is in `code/compute_worked_example.py`.

6.2 The decline: spot falls from $50 to $30

Merchant generators absorb the full shock: revenue falls from $1,500M to $900M, a loss of $600M borne by generator shareholders (Table 2). PPA revenue does not move; the $25/MWh difference between the contract price and the market price is a loss to the contract counterparty, $500M across the PPA volume, which the counterparty carries against the retail contracts it has written. CfD revenue does not move either: the generator nets the $60 strike, and the scheme's payment rises from $150M to $450M. Premium revenue does not move; the premium rises from $300M to $500M. Regulated revenue does not move, because no part of it is priced off the spot market.

The gap payments funded by consumers therefore rise from $450M to $950M. Spread over all consumption, the levy rises from $4.50/MWh to $9.50/MWh (Figure 1, top panel). The average consumer cost falls, but by $6.00/MWh rather than $20.00: only the 30 percent merchant share repriced. For a customer buying at the spot price, the levy claws back a quarter of the $20 saving ($5.00 of the $20.00). For the representative household on a fixed retail tariff, nothing happens to the energy line within the contract year; at the next levy reset its levy cost rises from $45 to $95 a year. Where large users are exempt from the levy, the same $950M falls on a 60 TWh base and the household's levy cost rises to $158.33 a year: the exemption does not reduce the gap, it reallocates it.

6.3 The increase: spot rises from $50 to $70, and the asymmetry

Merchant revenue rises to $2,100M. The two publicly backed designs now part company, and the difference is the paper's central mechanical point. The CfD payment reverses sign: at a $70 market price the strike of $60 is below the market, so generators pay $150M back into the scheme, and the combined levy turns negative, a net credit of $0.50/MWh to consumers. The premium does not reverse. It shrinks to $100M and stops at zero by construction; if the spot price rose above $80, the premium generator's revenue would rise with the market, $\max(P, 80)$, keeping the whole upside. Stated precisely: a two-way CfD fixes generator revenue at the strike in both directions; a one-way premium floors generator revenue at the strike and leaves it uncapped. Ontario's GA behaves like the CfD in aggregate, because it nets contract payments against actual market revenue, and the aggregate GA has printed negative in individual months (Section 7.8). A fixed feed-in tariff with system resale has the same aggregate property; a floored premium does not.

Timing separates the customer classes. A large user on an indexed contract gains or loses the full $20/MWh on its spot-exposed volume immediately: $1,000,000 a year on 50 GWh. The same user pays the levy change at the next reset, and in the exemption variant pays no levy at all. A household on a fixed tariff sees only the levy line within the year, and sees the energy line only when its retail contract or the regulated default tariff reprices. Table 4 collects the incidence and timing by channel.

6.4 The no-spot variant: a cost shock inside a regulated utility

Replace the spot shock with the shock a regulated system actually experiences. The regulated segment serves 25 TWh at an allowed cost of $58/MWh, and its generation and procurement costs rise by

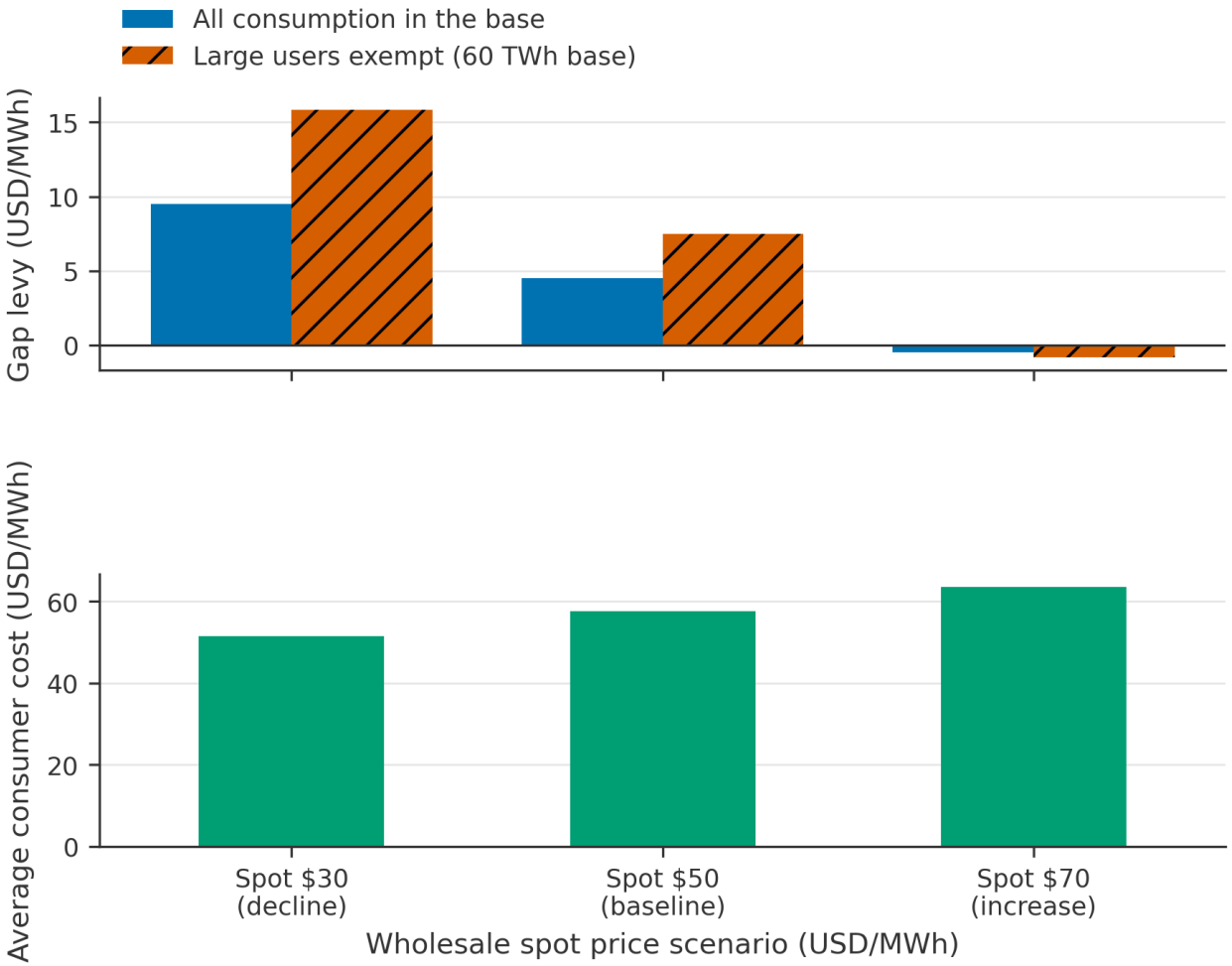


Figure 1: Worked example: the gap levy and the average consumer cost. Top panel: the levy that funds the publicly backed channels rises when the spot price falls and turns negative when the spot price rises above the CfD strike, under both levy bases. Bottom panel: the average consumer cost moves \$6.00/MWh against a \$20.00/MWh spot move. Source: Desk computation from `data/worked_example_assumptions.json` (Modelled, C). Note: Stylized market of Section 6.1; consumption held constant at 100 TWh.

\$10/MWh, an annual cost increase of \$250M (Modelled, C). There is no levy and no contract settlement. Two recovery regimes produce two different incidence paths for the identical cost.

Under a fuel adjustment clause with a one-quarter lag, customers pay the increase from the second quarter. The first quarter's \$62.5M is booked to a deferral account and recovered over the following year with a carrying cost of about \$1.6M at a 5 percent carrying rate: customers ultimately pay the full increase plus financing, and the utility's shareholders finance one quarter. Once effective, the increase costs the representative household \$100 a year and the large user \$500,000 a year.

Under a pure rate-case regime with the next case in 18 months, nothing reaches bills for a year and a half. The utility absorbs \$375M of cumulative margin over the interim. At the rate case the regulator decides whether the deferred costs were prudently incurred: allowed, they become a regulatory asset recovered from future customers, which is the Spanish and Indian device of Section 7; disallowed, they stay with shareholders. The choice between the two regimes is a choice about who finances the lag and who bears disallowance risk, not about whether the cost exists.

7 Market evidence

The subsections follow consumption order. For each market we record the channel mix with covered volumes where published, how the payment is determined, the recovery route, direction, reset timing by customer class, and the fall and rise trace. Figures carry the evidence grade of Section 4; Appendix A lists every load-bearing figure with its source and verification status. Three reading notes apply throughout. First, volumes are covered volumes under an arrangement, not country totals; Section 9 aggregates them. Second, where a market runs several arrangements, the dominant revenue channel for a generator is stated before the overlays. Third, monetary magnitudes are given in local currency as published; no conversion is imposed because the paper's comparisons are of mechanism, not of level.

7.1 China: a capped market, a frozen surcharge, and a two-way turn

China is the largest system in the world (9.85 trillion kWh consumed in 2024 on the National Energy Administration's measure (Observed, A); 10,066 TWh on Ember's demand concept) and the clearest case of a gap that was socialized by not collecting it. Three arrangements coexist.

Most industrial and commercial volume now trades. Market-traded electricity was 6.2 trillion kWh in 2024, 63 percent of consumption (Observed, A), but it trades inside a regulated frame: since October 2021, market prices for coal-fired power float within plus or minus 20 percent of provincial coal benchmark tariffs, with the ceiling removed for energy-intensive enterprises, while residential and agricultural users stay on regulated catalogue tariffs at stable prices (Observed, A). This is Channel P inside a price band, with the band itself a policy variable.

Legacy renewables were paid feed-in tariffs above the benchmark, with the premium funded by a renewable surcharge on sales paid into the Renewable Energy Development Fund. The surcharge was last raised in 2016, to RMB 0.019/kWh (Observed, B), and was not raised as obligations grew. The difference accumulated as a fund deficit and as multi-year payment arrears to generators: a cumulative deficit of RMB 821B for 2012–2024 is reported by Bruegel (Working Paper 15/2026), attributed there to Ministry of Finance data (Observed, B); the ministry document itself was not read in this paper's verification (Appendix A). China's deferral is the Spanish device at larger scale: the charge stayed fixed, so the gap became a claim on the fund rather than a bill line.

The traced shocks follow from the design. When coal and market prices fell, household bills did not move (catalogue tariffs) and the FIT gap widened silently into arrears. When prices spiked in 2021, the state widened the band to ±20 percent and let energy-intensive users pay uncapped prices, socializing the shock selectively onto commercial and industrial users while households were shielded; coal generators selling below cost curtailed output and shortages followed. Since 1 June 2025, under NDRC/NEA Document 136, new wind and solar sell at market prices and a provincially auctioned mechanism price settles the difference outside the market, two-way in design, with settlement costs carried in local system operation expenses (Observed, B). China is thus building the Ontario mechanism for its new renewables while its legacy gap remains an arrears stock.

7.2 The United States: three systems in one country

US retail sales were 3,975 TWh in 2024 (Observed, A). The country contains three distinct answers to the paper's question, and any single US number is a weighted average of them.

Regulated territories. Across the Southeast, Southwest and Northwest, vertically integrated

utilities own generation and recover approved costs plus an allowed return in commission-set base rates (FERC's characterization of traditional markets) (Observed, A). Between rate cases, fuel and purchased-power costs pass through fuel adjustment clauses: a per-kWh charge or credit equal to actual cost against the base-period cost embedded in rates, reset periodically (monthly in the Kentucky rule examined for this paper, 807 KAR 5:056) and reviewed for prudence after the fact. There is no market-price gap to compute because the utility's fleet has no market price; the pass-through is of actual cost. A fuel price fall becomes a smaller FAC charge or a credit at the next reset; a rise becomes a surcharge on the same clock. Both households and large users ride the same per-kWh rider, so the fall reaches both, smoothed by the reset calendar.

Restructured markets. Two-thirds of US load is served in ISO/RTO regions with bid-based energy markets (Observed, A). Energy there is merchant, and capacity constructs in PJM, ISO-NE, NYISO, MISO and CAISO pay for availability rather than the energy-price gap, so they are not classified as gap mechanisms here. The gap layer in these footprints is state-imposed. New York's Clean Energy Standard obliges load-serving entities to buy renewable energy certificates, offshore wind ORECs and nuclear zero-emission credits from NYSERDA in proportion to load; utilities recover the cost volumetrically, dividing annual cost by projected kWh sales and reconciling annually (tariff Rule 39 of NYSEG/RG&E) (Observed, A). The ZEC programme alone covers up to 27,618 GWh a year (Observed, A). New Jersey funds offshore wind through a non-bypassable per-kWh OREC surcharge set annually by the Board of Public Utilities as forecast requirement over estimated sales, with market revenues credited back (NJ Administrative Code §14:8-6.6) (Observed, A); no New Jersey project is yet operating, so the charge exists as a formula, not a collected rate. These are Ontario-form charges on the contracted slice only: when wholesale prices fall, the net cost of the fixed-price contracts rises and the surcharge follows at its next annual setting, while the merchant remainder of the bill falls. Per-kWh CES surcharge rates could not be primary-verified (Appendix A).

ERCOT. Texas is the large-scale merchant exception: an energy-only market with no capacity market and a system-wide offer cap of $5,000/MWh, serving about 90 percent of Texas load (Observed, A). ERCOT generation was about 482 TWh in 2024 (Observed, B). When prices fall there is no levy to rise; generator revenue falls one-for-one and scarcity rents disappear. When prices spike to the cap, exposed buyers pay in real time and no socialized true-up follows. EIA's supplier-type split is a useful bound on how much of the US is even exposed to retail competition: energy-only (competitive) suppliers served 14.9 percent of 2024 retail sales, full-service providers 85.1 percent (Reconstructed, A), though full-service sales include default service in restructured states and are not a market-design split.

7.3 The European Union: five member states and a common rule

7.3.1 The EU-level rule

Regulation (EU) 2024/1747, the 2024 electricity market design reform, inserted Article 19d into Regulation (EU) 2019/943: from 17 July 2027 (17 July 2029 for hybrid offshore installations), direct price support for new wind, solar, geothermal, reservoir-less hydro and nuclear generation must take the form of two-way contracts for difference or an equivalent scheme (Observed, A). Below the strike the generator is topped up; above it, the excess is clawed back. The Regulation fixes the direction of the settlement and leaves the payer (levy, budget or a national counterparty) to national design. The member states below show the legacy stock the rule will gradually replace: mostly one-way instruments, funded through levies that several states have already moved onto their budgets.

7.3.2 Germany: the EEG, from levy to budget

Germany's EEG pays small installations a fixed feed-in tariff for 20 years and larger, directly marketed installations a sliding market premium: the applicable reference value minus the market value of the electricity. The design is one-way; the premium floors at zero and there is no clawback above the reference value. In 2024, 238.0 TWh was entitled to EEG payments and €18.5B was paid out, an average of 8.0 ct/kWh (Observed, A).

Until mid-2022 the gap was funded by the EEG-Umlage, a per-kWh levy calculated in terms as EEG payments minus exchange-sale revenues, divided over consumption: the Ontario GA formula, set annually each October on forecasts. The final levy was 3.723 ct/kWh; from 1 July 2022 it is zero, and EEG payments are financed from the federal Climate and Transformation Fund (Observed, A). The 2017 peak levy of 6.88 ct/kWh is widely reported but rests on secondary sources in this paper's verification (Observed, B). Energy-intensive manufacturers were partly shielded throughout by the Besondere Ausgleichsregelung, administered by BAFA, which capped the surcharge they could be charged; the cost of the cap fell on non-privileged consumers.

A price fall widens the premium and raises EEG payments; before 2022 that fed mechanically into the following year's levy, and since 2022 it lands on the federal budget instead. A price rise shrinks the premium toward zero and stops there: the payer pays less, nobody pays back, and generators keep revenue above the reference value.

7.3.3 France: purchase obligations, budget compensation, and the end of ARENH

France supports renewables through the obligation d'achat (EDF and local distributors must buy supported output at a regulated tariff) and the complément de rémunération, a premium over the market reference price. Both are compensated as public-service charges: the regulator CRE assesses the charges annually and the State compensates the obligated buyers. The formula is symmetric (support cost minus the market sale revenue of the supported output), and in the first half of 2023 it produced negative compensation of €2.7B on EDF's perimeter, because supported renewables sold for more than their support cost (Observed, A). Supported volume in 2024 was about 63 TWh, of which 15.7 TWh under the premium scheme (Reconstructed, A). Funding runs through the State budget; the electricity excise (33.70 €/MWh for households before August 2025, 29.98 €/MWh since) is general budget revenue, not a gap-tracking levy (Observed, A).

Alongside support, ARENH obliged EDF to sell up to 100 TWh a year of historic nuclear output to rival suppliers at a regulated €42/MWh (Observed, B); at high market prices this transferred the nuclear rent from EDF to suppliers and their customers. CRE confirms the framework ended on 31 December 2025 (Observed, A). Regulated sales tariffs remain for small consumers (up to 36 kVA); CRE reports about 52 percent of sites on regulated tariffs in March 2026 against 77 percent of volume at market prices, which is the French version of the class split: households are insulated by tariff regulation, large users face the market.

7.3.4 Spain: the tariff deficit, or the charge deferred

Spain is the limiting case of suppressing the gap charge. Feed-in tariffs for the régimen especial were paid as regulated system costs, but governments held access tariffs below recognized costs. The shortfall became the tariff deficit: a receivable owed to the system, financed first by the large utilities and then securitized through FADE, a state-guaranteed fund repaid from future access charges. The accumulated deficit passed €30B by end-2012 on the CNE-based accounting used by Espinosa (2013),

with about €22B outstanding (Observed, B); a higher cumulative figure (€35.6B) circulates on a different definition, and no CNMC primary table was read in this paper's verification, so the deficit total is carried as secondary-only (Appendix A). Either way, the structure is not in dispute: the GA was not avoided, it was converted into debt and collected from future consumers with interest.

The 2013 reform (Law 24/2013) repealed the feed-in regime and substituted a specific remuneration: installations sell at market and receive a regulated top-up calibrated so a standard plant earns a reasonable return (7.398 percent pre-tax for the first regulatory period) (Observed, B). The regulated bill is now split between peajes (network tolls, set by the CNMC) and cargos (system charges set by the government, carrying support costs and deficit annuities) (Observed, A). Small consumers on the PVPC tariff are indexed to the wholesale price hourly, so the class incidence has inverted relative to the deficit era: households now bear spot exposure directly, while the legacy gap sits in cargos.

7.3.5 Italy: ASOS and the cost of exemptions

Italy funds support (Conto Energia feed-in tariffs, CIP6, and auctioned two-way CfDs under the FER decrees for newer plant, administered by GSE) through ASOS, the largest component of the general system charges on bills, reset by ARERA quarterly. ARERA's Q3 2025 split: ASOS is 94.75 percent of general system charges, of which 70.58 percentage points are renewables incentives and 24.17 percentage points are the concessions granted to energy-intensive enterprises (Observed, A). The exemption is not outside the charge; it is a line inside it, paid by non-exempt customers. System charges were 3.13 eurocent/kWh, 10.1 percent of a typical protected customer's bill, in Q3 2025. Support costs in 2025 were about €9.3B on roughly 50.2 TWh of incentivized volume (Observed, A). Fixed legacy tariffs do not respond to price falls, so ASOS declines slowly when market prices rise; the two-way FER contracts pay back above strike, but they are a small part of the stock.

7.3.6 Poland: itemized charges inside a regulated tariff

Poland's household tariffs are approved by the regulator URE, and bills carry itemized statutory charges inside the distribution tariff: the RES (OZE) charge funding renewables support, a cogeneration charge, and a capacity levy funding the capacity market (PLN 17.18 per month for a household consuming 1.2–2.8 MWh a year; per-kWh for other users). For 2026 URE approved a RES charge of PLN 7.3/MWh, up from PLN 3.50 in 2025 and zero before that, and a cogeneration charge of PLN 3/MWh (Observed, A). The charges are set annually in the tariff process; they do not track market prices month to month, so the Polish levy adjusts to support costs with a lag of a year or more. During the price spike, households were instead insulated by a statutory price freeze at PLN 500/MWh through September 2025, with suppliers compensated from public funds (the funding route and amounts rest on secondary reporting; Appendix A) (Observed, B). For 2026 the freeze has lapsed and the approved household tariff (PLN 495.16/MWh) sits about 1 percent below the old frozen level, because contract and wholesale prices fell (Observed, A).

7.4 India: the gap inside discom tariffs, deferred as regulatory assets

India has no national gap levy. Generation is bought predominantly under long-term PPAs by state distribution companies (discoms): cost-plus regulated tariffs for thermal plant, auctioned fixed tariffs for solar and wind, often intermediated by SECI or NTPC. The short-term market is small: power exchanges cleared 143.75 BU in FY2024-25, 7.9 percent of gross generation of 1,829.69 TWh, and all

short-term transactions were 13.03 percent, so about 87 percent of generation sits under long-term arrangements (Observed, B).

The gap appears as the difference between the average cost of supply and average revenue (the ACS–ARR gap, Rs 0.19/kWh in FY2023-24 on a subsidy-received basis (Observed, A)) and is disposed of three ways: state budget subsidy (20.21 percent of discom revenue in FY2023-24, concentrated on agriculture and households (Observed, A)), cross-subsidy from industrial and commercial tariffs set above cost, and deferral. Deferral is the regulatory asset: a state commission allows the discom to book revenue recoverable from future tariffs rather than raise tariffs now. PFC's state-level figures put Delhi's stock at Rs 27,200 crore, Tamil Nadu's at Rs 89,375 crore and Rajasthan's at Rs 49,842 crore at end-FY2024 (Observed, A); six jurisdictions alone sum to about Rs 1.77 lakh crore. No PFC-published all-India total was found in this paper's verification: Rs 1.74 lakh crore is the figure used in the Supreme Court's August 2025 judgment on regulatory assets, and about Rs 3 lakh crore is ICRA's estimate, both secondary here (Observed, B). Discom borrowings of Rs 7,52,677 crore (Observed, A) are the financing counterpart of the same deferral.

The shock trace is one-way and slow. An exchange price fall touches the 8 percent of energy bought on exchanges and barely moves average purchase cost. A fuel or purchase-cost rise presents the commission with the Spanish choice: raise tariffs now, or book the shortfall as a regulatory asset and recover it, with carrying cost, from future consumers. Regulatory assets only ever move in one direction; there is no market-price clawback anywhere in the Indian design. Large users respond by exiting where open access and captive generation allow, shrinking the cross-subsidy base that funds the system.

7.5 Japan: the surcharge that is net of the spot price

Japan's feed-in scheme, since 2012 a feed-in tariff and since 2022 a feed-in premium for larger projects, pays fixed prices (or a fixed premium over market revenue) for 20 years. The funding charge is the cleanest published gap number anywhere: METI sets the Renewable Energy Surcharge each year as forecast purchase costs minus avoidable costs, divided by assumed electricity sales. For FY2025 the published components are purchase costs of ¥4.8540 trillion and assumed sales of 770.8 TWh, giving ¥3.98/kWh, or ¥1,592 a month for a 400 kWh household (Reconstructed, A). Because avoidable costs are indexed to actual JEPX spot prices, the surcharge is a sampled, one-year-lagged version of the Ontario formula. The collapse and rebound in Figure 2 is the mechanism in action: spot spikes in FY2022 raised FY2023 avoidable costs to ¥3.6353 trillion (Observed, A) and cut the FY2023 surcharge to ¥1.40/kWh (Observed, B); as wholesale prices normalized, the surcharge returned to ¥3.49 (FY2024) and ¥3.98 (FY2025) (Observed, A).

The rest of the Japanese consumer price is mostly regulated or default tariffs set by ten legacy utilities plus an automatic monthly fuel cost adjustment pass-through, an FAC operating inside a retail system rather than a wholesale one. Large users on market-linked retail contracts take spot exposure directly. The FIT fleet's purchased volume in TWh is not published in METI's English releases (Appendix A), which is why Japan contributes to the unknown share in Section 9 despite having the best-documented charge.

7.6 Brazil: regulated auctions and a free market share the gap charge

Brazil runs two contracting environments side by side. In the regulated environment (ACR), distribution companies buy through government auctions, including the PROINFA legacy contracts, and pass costs through annual regulated tariff adjustments; auctions awarded about 61 percent of 2024

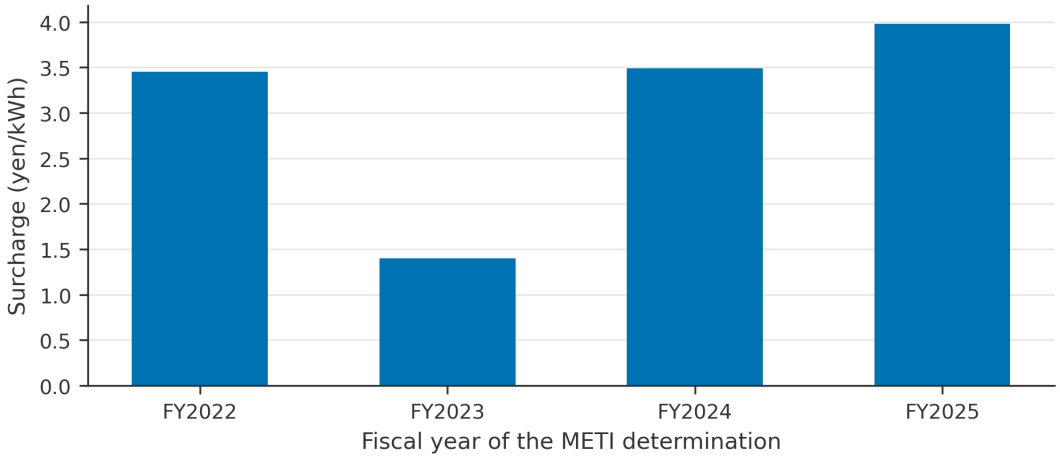


Figure 2: Japan's Renewable Energy Surcharge (yen/kWh), METI annual determinations, FY2022–FY2025. The FY2023 collapse reflects the FY2022 spot-price spike raising avoidable costs; the rebound follows as wholesale prices normalized. FY2023 rate: (Observed, B); other years (Observed, A).

consumption on this route (Observed, A). In the free environment (ACL), large consumers contract bilaterally; this was 39 percent of consumption in 2024, rising toward 42 percent in 2025 (Observed, A). The spot price (PLD) settles differences in both.

The gap charge is the Conta de Desenvolvimento Energético (CDE), a sector fund collected inside distribution tariffs that pays, among other items, fuel costs for isolated systems (the CCC line), the social tariff for low-income households, universalization and distributed-generation compensation. ANEEL's 2025 budget determination (Resolução Homologatória 3.484/2025) sets the CDE at R$49.227B, of which consumers fund R$46.886B through the CDE-USO and CDE-GD quotas (Observed, A). Two features distinguish it from Ontario. First, CDE flows are predominantly transfers and policy costs, of which only part is a market-price gap on contracts; the fund is broader than the GA. Second, hydro risk, the system's main price risk, is socialized separately through the generation scaling factor mechanism, under which shortfalls of the hydro fleet against its assured energy are shared across all hydro generators and financially hedged through the energy reallocation mechanism.

Hydrology is Brazil's traced shock: in a dry sequence the PLD rises, ACL contracts reprice at renewal, ACR tariffs absorb auction and spot exposure at the annual reset, and tariff flags add an immediate per-kWh surcharge in scarce months, so Brazilian households see scarcity within the month through the flags and within the year through the reset, while Ontario households see it only in the smoothed GA.

7.7 South Korea: a single-buyer pool and a certificates overlay

South Korea sells almost all generation into KPX's cost-based pool, with KEPCO as the dominant buyer and retailer. The System Marginal Price is computed from the variable costs of the marginal generator and averaged KRW 167.0/kWh in 2023, 128.3 in 2024 and 112.7 in 2025 (Observed, B). Because KEPCO buys at SMP and sells at regulated tariffs that were frozen for households through the recent cycle, the Korean gap materialized as KEPCO operating losses and debt rather than as a bill line: the Chinese and Spanish deferral in corporate form. When the government did move, it moved asymmetrically: in October 2024 industrial tariffs rose 9.7 percent while household tariffs stayed frozen (Observed, B).

On top of the pool sits the Renewable Portfolio Standard: generators of 500 MW or more must supply a rising share as renewables (13.5 percent in 2024, 14 percent in 2025, 15 percent in 2026, rising to 25 percent by 2030) (Observed, A), met by their own build or by buying renewable energy certificates at around KRW 70.2/kWh in 2024 (Observed, B), on about 63.8 TWh of RPS generation (Observed, B). Compliance costs enter KEPCO's cost base and reappear in bills only as the published climate and environment charge of KRW 9/kWh, plus the capped fuel adjustment (±KRW 5/kWh). A spot-price fall therefore reduces the pool deficit invisibly; households see none of it until a tariff decision converts it.

7.8 Canada: Ontario's Global Adjustment, and the regulated rest

Ontario is the reference case because it publishes the mechanism monthly. The IESO clears a wholesale market in which contracted and regulated generators are paid the market price, and then settles the difference between their contract or regulated rates and the market centrally. The residual is the Global Adjustment, described by the IESO as covering "the cost of building new energy infrastructure, maintaining existing resources and delivering government conservation programs" (Observed, A). Its scale over the decade to September 2026: GA payments of C$94.5B against C$47.1B of commodity value, C$2.01 of GA for every C$1 of energy (Figure 3; (Reconstructed, A)). In 2020 the ratio reached 7.61, when GA of C$14.0B sat on a commodity base of C$1.8B; by 2025 it had fallen to 0.55 as market prices firmed.

Incidence inside the GA is two-part. Class A customers (large users, generally above 1 MW, and manufacturers above 500 kW) pay a share of the total GA proportional to their contribution to the five highest provincial peak hours (the Peak Demand Factor), so their GA is a capacity-like charge they can reduce by curtailing on peaks, a feature with no counterpart in any levy examined here. Class B (households, farms, small business) pays the remainder at a monthly per-kWh rate; for households it is embedded in the Ontario Energy Board's Regulated Price Plan rates, reset each May and November, so a GA movement reaches a household bill months later and smoothed (Observed, A). The estimate ladder is monthly: a first estimate on the last business day of the prior month, a second estimate at month end, and the actual on the tenth business day of the following month (IESO Market Manual 5.6) (Observed, A). The charge is two-way in practice: GA has been negative in 18 months since 2005, most sharply in January 2026 at minus C$494M, when high market prices (a weighted average of C$139.63/MWh) made contracted generators net payers (Reconstructed, A). About 85 percent of the cost of non-hydro renewable contracts was moved off the rate base onto the provincial tax base from 1 January 2021, splitting the payer between ratepayers and taxpayers within one scheme (Observed, A).

The rest of Canada is the no-spot variant at national scale: Crown or investor-owned cost-of-service utilities in Quebec, British Columbia, Manitoba, Saskatchewan and New Brunswick recover generation costs in regulated rates, and no other province publishes a GA-style itemized charge (Observed, B). Alberta is the merchant exception: an energy-only market with internal load of 88.8 TWh in 2024 (Observed, A), where a price fall is borne by generators directly, as in ERCOT but with a $1,000/MWh administrative cap.

7.9 United Kingdom: a two-way CfD with a floored levy

British low-carbon support runs through three levy-funded schemes. The contract for difference (CfD), administered by the government-owned Low Carbon Contracts Company (LCCC), is two-way by contract: LCCC pays the generator the strike price minus the market reference price, and the

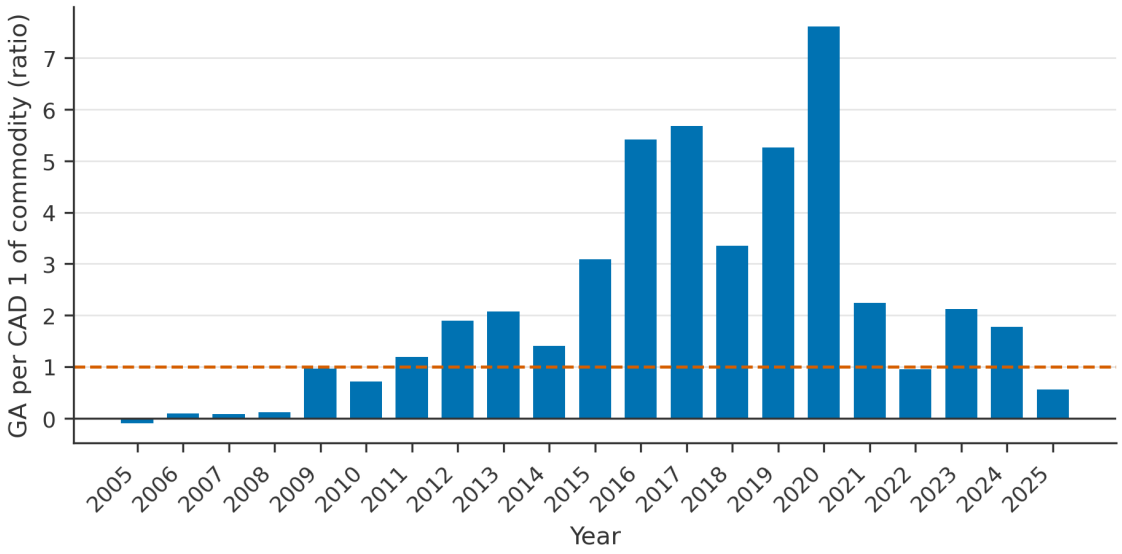


Figure 3: Ontario: Global Adjustment per $1 of wholesale commodity value, by year, 2005–2025. The dashed line marks parity. The ratio is highest when market prices are lowest, the inverse signature of a contract-settlement charge. Source: (Reconstructed, A) from IESO settlement data.

generator pays LCCC when the reference price exceeds the strike. Suppliers fund LCCC through the Supplier Obligation, collected as a daily Interim Levy Rate (£/MWh) set a quarter ahead, with reconciliation after each quarter. The legacy Renewables Obligation and the small-scale Feed-in Tariff are one-way and levy-funded.

The floor on the levy is the British asymmetry, and it is the mirror image of the premium floor in Section 6. The Interim Levy Rate cannot be set below £0/MWh. When wholesale prices rose above strikes from September 2021, the rate went to zero for the first time and CfD generators became net payers: in Q4 2021 they paid £133.7M net, and after reconciliation £39.2M flowed back to suppliers (Observed, A). The consumer gain from the payback arrived through a zero levy and a quarterly reconciliation, not through a negative charge, and it was small next to the wholesale cost itself. In a price fall the mechanism runs the other way within one quarter: LCCC forecasts higher difference payments, the levy is set above zero, and bills fund the top-ups. CfD generation in 2023/24 was 22.2 TWh (Observed, A), against total demand of 318.7 TWh in 2024 (Observed, A), so the two-way contract covers a single-digit share of volume; the rest of the British bill's exposure to the wholesale price runs through supplier hedging and the regulated price cap, which resets quarterly.

From April 2026, 75 percent of the *domestic* cost of the legacy Renewables Obligation is met by the Exchequer for three years (£7B in total), the rest of the RO staying on bills (Observed, A). It is the same payer substitution as Germany's: the settlement continues, the levy line shrinks, and taxation replaces it.

7.10 Australia: merchant energy, certificates, and a taxpayer collar

The National Electricity Market is the deepest merchant market in the sample: generators are paid the regional spot price, retailers hedge privately, and no gap levy exists for energy. NEM operational consumption was 178 TWh in 2024-25 (Observed, A). The publicly backed layer has two forms. The Large-scale Renewable Energy Target creates certificates (LGCs) that liable retailers must surrender

against a statutory target of 33,000 GWh a year, with the Renewable Power Percentage set at 17.91 percent for 2025 (Observed, A): a certificate scheme whose cost passes into retail prices without ever appearing as a named charge. The Capacity Investment Scheme, by contrast, is taxpayer-funded: under its tender guidelines the Commonwealth tops up 90 percent of a project's shortfall below an annual revenue floor and takes back 50 percent of revenue above an annual ceiling, subject to an annual support cap and a term of up to 15 years (Observed, A). The scheme was enlarged in July 2025 from 32 GW to 40 GW of capacity (Observed, A).

Australia is therefore the control case for the funding question: the same gap payment that Ontario funds from ratepayers and Germany funded from a levy until 2022, Australia funds from general taxation, and the partial clawback (50 percent, not 100) leaves the generator with half of any upside, a deliberate asymmetry. A spot-price fall raises CIS top-ups (taxpayers) and lowers LGC scarcity value only if certificate supply loosens; household bills see merchant price falls through retail competition within months, faster than any levy-funded system in this paper.

7.11 The Nordic market: merchant by default

The Nordic market is the control case. Generation connected to Nord Pool (Norway, Sweden, Finland, Denmark) sells predominantly at zonal spot prices and bears price risk; there is no feed-in tariff and no gap levy. The one socialized overlay was the Swedish-Norwegian electricity certificate scheme (2012): producers received one certificate per MWh for up to 15 years, and suppliers and certain consumers surrendered certificates against a quota. The certificate price is set by certificate supply and demand, so the consumer cost (quota times certificate price, embedded in retail prices) does not rise mechanically when spot prices fall. The joint target of 28.4 TWh of new renewable production was reached in 2019, the scheme closed to new entrants at the end of 2021, and it ends in 2035 (Observed, A).

The price-shock trace is therefore the shortest in this paper. When Nordic prices fall, merchant generators absorb the loss and consumers on spot-linked contracts gain at once; there is no counterparty to bill. When prices rise, generators keep the full upside: the 2021–22 Nordic price spike accrued to hydro and nuclear owners and was politically contested precisely because no clawback existed. The certificate overlay neither amplified nor offset the shock materially by then, because certificate prices had already collapsed toward administrative cost. The Nordic market shows what the rest of the paper's mechanisms are deviations from: full spot exposure, with risk allocated by private hedging alone.

7.12 South Africa: building the gap charge into the new market

South Africa's legacy system is a vertically integrated Eskom selling under NERSA-regulated tariffs set through multi-year price determinations, with allowable revenues for generation, transmission, distribution and legacy items published in NERSA's reasons for decision (Observed, A). Independent power producers under the REIPPPP programme hold 20-year rand-denominated PPAs with Eskom as buyer (Observed, A), so their cost enters the regulated revenue requirement and is recovered in tariffs, Ontario's outcome without an itemized charge.

The reform now separates the two. Under the South African Wholesale Electricity Market being stood up by the National Transmission Company, legacy and vesting contracts are to be hedged outside the market and "the difference between the System Marginal Price and the hedge in the Legacy Contracts and the Vesting Contracts shall be covered by a legacy charge paid by all consumers", calculated annually and approved by NERSA (NERSA Wholesale Electricity Pricing Methodology consultation;

NTCSA draft CPA Conduct Rules) (Observed, A). South Africa is adopting the Ontario formula explicitly, before the legacy stock is closed, and the consultation status of both documents is flagged in Appendix A.

7.13 Mexico: the gap absorbed by the Treasury

Mexico's wholesale market (MEM) coexists with a dominant state utility: under the 2025-2030 Energy Sector Programme the State must inject at least 54 percent of the energy in the national system, and CFE holds at least 54 percent in mixed-investment projects under the October 2025 regulations (Observed, A). Legacy self-supply and independent-producer contracts with CFE predate the market; their above-market cost sits inside CFE's regulated revenue rather than in any itemized charge, and could not be quantified from primary sources in this paper's verification (Appendix A).

The Mexican gap is fiscal. Residential and agricultural tariffs are held below supply cost and the difference is a budgeted transfer: the subsidy framework is published in the SHCP transfer rules for CFE Basic Supply (Observed, A), and CFE's director general put the 2025 subsidy at MXN 84,805M, reaching 89 percent of users and covering 47 percent of the average bill (Observed, B). When generation costs rise, the first absorber is the federal budget; tariffs are adjusted by policy decision, not formula. Mexico is the limiting case of taxpayer funding: no levy, no certificates, no itemization, and incidence determined by the tax system rather than by electricity consumption.

7.14 The Philippines: FIT-All and the collection holiday

The Philippines' feed-in tariff is funded by the FIT-All, a uniform peso-per-kWh charge on all grid consumers, collected by distribution utilities and paid into a fund administered by TransCo, which pays eligible renewable generators the difference between their FIT rate and market revenue. Total generation in 2024 was 126,941 GWh (Observed, A); the FIT fleet is small, 84 plants and 1,707.63 MW (Observed, B).

The verified rate path corrects the usual description of this scheme (Observed, B). The FIT-All was not zero in 2022–2024: the 2022 rate was ₱0.0364/kWh, collection (not the scheme) was suspended from December 2022 to January 2024 because the fund was in surplus, and the rate then rose as WESM prices fell and the differential widened: ₱0.0838/kWh from June 2024, ₱0.1189 from March 2025, and ₱0.2073/kWh from November 2025, the last to settle a ₱19.06B differential. The sequence is the two-way mechanism observed from the payer's side: a spot-price fall of 2023–2025 vintage produced a rise of more than fivefold in the per-kWh charge within two years, delayed by the collection holiday while the accumulated fund absorbed it. Rates rest on Energy Regulatory Commission decisions reported in trade and official-adjacent coverage; the decision texts themselves were not read in this paper's verification and the flag is recorded in Appendix A.

8 Summary of mechanisms

Table 5 compresses Section 7 into the five incidence variables of Section 5. Two patterns are visible before any aggregation. First, the payer is almost always the electricity consumer: of the eighteen rows, only Germany's EEG (budget-funded since July 2022), Australia's Capacity Investment Scheme and Mexico's residential subsidy place the gap on taxpayers in the first instance, and each of those systems arrived at budget funding after a period of levy funding or as a deliberate substitute for it. Second, two-way designs cluster where the charge is formula-set by an administrator (Ontario, the

UK, Japan, the Philippines, South Africa's proposed legacy charge), while one-way designs cluster where the payment is a tariff or premium set in advance (legacy feed-in systems, India's regulatory assets, China's arrears). Direction is a property of the settlement formula, not of the technology supported.

Table 1: Markets examined: consumption and data sources.

Market	Consumption 2024 (TWh)	Channels	Main sources
China	10,066	P, R	Ember Global Electricity Review 2025
United States	4,401	G, R, S	Ember Global Electricity Review 2025
EU-27	2,727	G	Ember Global Electricity Review 2025
India	2,054	R, S	Ember Global Electricity Review 2025
Japan	1,027	none published	Ember Global Electricity Review 2025 (Set A vintage)
Brazil	775	P, R	Ember Global Electricity Review 2025 (Set A vintage)
South Korea	628	G, R	Ember Global Electricity Review 2025 (Set A vintage)
Canada	654	G, R, S	Ember Global Electricity Review 2025 (Set A vintage)
United Kingdom	319	G	DESNZ Digest of UK Energy Statistics (final demand 318.7 TWh)
Australia	265	G, S	Ember Global Electricity Review 2025 (Set A vintage)
Nordics	390	S	Ember Global Electricity Review 2025 (Set A vintage; Sweden 137 verified)
South Africa	210	R	Secondary estimate (not confirmed in Ember sources read); flagged
Mexico	390	R	Ember Global Electricity Review 2025 (Set A vintage)
Philippines	127	G	Philippines DOE power statistics (generation 126.9 TWh prorata)
Russia	1,178	R	Ember Global Electricity Review 2025 (Set A vintage)

Notes: Consumption is electricity demand as reported by Ember (Global Electricity Review, 2024). Channels are those with published covered volumes used in Section 9; a market may operate a channel whose volume is not published, in which case the volume sits in the unknown share. Source: Ember; regulator and operator publications per market (Appendix A).

Table 2: Worked example: generator revenue and gap payments by scenario. Stylized market, 100 TWh, consumption held constant. USD millions per year.

	Volume (TWh)	Spot $30 (decline)	Spot $50 (baseline)	Spot $70 (increase)
Merchant (spot)	30	900	1,500	2,100
Private PPA, $55/MWh	20	1,100	1,100	1,100
Two-way CfD, strike $60/MWh	15	900	900	900
One-way premium, floor $80/MWh	10	800	800	800
Regulated cost of service, $58/MWh	25	1,450	1,450	1,450
Total generator revenue	100	5,150	5,750	6,350
Gap payments funded by the levy (CfD + premium)		950	450	-50

Notes: The two-way CfD pays the generator the strike in every scenario; its gap payment is negative at a $70 spot price, i.e. generators pay levy payers back. The one-way premium is floored at zero and never reverses. The accounting identity holds in every scenario: consumer outlay (supplier procurement + regulated tariffs + gap payments) equals total generator revenue. Source: Desk computation from `data/worked_example_assumptions.json` (Modelled, C).

Table 3: Worked example: what reaches the bill. USD per MWh unless stated.

	Spot $30	Spot $50	Spot $70
Gap levy, all consumption in the base	9.50	4.50	-0.50
Gap levy, large users exempt (60 TWh base)	15.83	7.50	-0.83
Average consumer cost, all channels	51.50	57.50	63.50
Household levy cost, USD/year (10 MWh, full base)	95.00	45.00	-5.00
Household levy cost, USD/year (large users exempt)	158.33	75.00	-8.33

Notes: A $20/MWh spot move changes the average consumer cost by $6.00/MWh in either direction (pass-through coefficient 0.30), because only the 30 percent merchant share reprices in year one. The levy moves against the spot price: it rises $5.00/MWh in the decline and falls $5.00/MWh in the increase. Source: Desk computation from `data/worked_example_assumptions.json` (Modelled, C).

Table 4: Worked example: incidence and timing by channel

Channel	Price risk borne by	Household sees it	Large user sees it	Reverses?
Merchant	Generator, then indexed buyers	At tariff reset	Immediately	n/a
Private PPA	Contract counterparty	At contract renewal	At contract renewal	No
Two-way CfD	Levy payers	At levy reset	At levy reset, or never if exempt	Yes
One-way premium	Levy payers (downside only)	At levy reset	At levy reset, or never if exempt	No
Regulated	Customers, with a lag	At FAC or rate case	At FAC or rate case	Only via true-up

Notes: "At levy reset" means the next scheduled reset of the scheme charge: monthly in Ontario, quarterly for the British CfD levy, annually for the Japanese FIT surcharge. Source: Desk framework, Section 5 (Modelled, C).

Table 5: Market-by-market summary: channels, schemes, payer, direction, reset lag, and incidence by customer class.

Market	Dominant channels	Named schemes	Payer	Direction	Reset lag	Household vs large-user incidence
China	P (banded), R	Coal benchmark bands; REDF surcharge; Document 136 mechanism price	Consumers; fund arrears	One-way legacy; two-way for new renewables	Annual tariff decisions	Households shielded on catalogue tariffs; energy-intensive users uncapped above the band
United States	S, P, R by region	Fuel adjustment clauses; NY CES (REC/ZEC/OREC); NJ OREC charge	Consumers	One-way (charges floored at zero in practice)	FAC monthly to quarterly; CES annual	Same per-kWh riders for both classes in regulated states; large users exit via retail choice and PPAs where allowed
Germany	S, P, G	EEG market premium; grid charges; BesAR caps	Taxpayers (EEG since 2022); consumers (grid)	Two-way by formula, settled against market value	Monthly premium settlement; charges annual	Capped industrial users shift the grid-charge share to households and small business
France	S, P, G	ARENH (ended 2025); complément de rémunération; excise	Consumers; EDF balance sheet	Two-way (CSPE compensation went negative in 2023)	Annual	Households on regulated sales tariffs see smoothed excise and tariff shield decisions
Spain	S, P, G	Regulated PVPC; historical tariff deficit; peajes and cargos	Consumers, deferred via deficit securitization	One-way with deferral	Semi-annual deficit true-ups historically	PVPC households took spot exposure directly until shield measures; industrial users contract out
Italy	S, P, G	ASOS component of general system charges	Consumers	One-way charges; concessions inside ASOS	Quarterly (ARERA)	Energy-intensive concessions (24.17 points of ASOS) are paid by other consumers
Poland	R, S, G	RES charge; cogeneration charge; capacity levy; tariff freeze	Consumers; budget for the freeze	One-way	Annual (URE)	Households protected by the freeze; capacity levy banded by annual consumption

Market	Dominant channels	Named schemes	Payer	Direction	Reset lag	Household vs large-user incidence
India	R, P	Long-term PPAs; regulatory assets; RPO/RCO	Consumers (deferred); state budgets	One-way; deferral one-directional	Annual tariff orders, often delayed	Households and agriculture subsidized; industry pays above cost and exits via open access
Japan	R, G	FIT/FIP surcharge; fuel cost adjustment	Consumers	Two-way netting against avoidable cost	Annual surcharge; monthly FAC	Uniform per-kWh surcharge; large users partly exempt via reduction schemes (secondary evidence)
Brazil	R (ACR), P (ACL)	CDE fund; PROINFA; tariff flags	Consumers	Mostly transfers; flags two-way with scarcity	Annual reset; flags monthly	Social tariff inside CDE shields low-income households; ACL users self-insure
South Korea	R, G	Cost-based pool; RPS certificates; climate charge	Consumers; KEPCO balance sheet	One-way charges; pool deficit deferred	Tariff decisions ad hoc	Households frozen while industry took the October 2024 increase
Canada (Ontario)	S energy, G settlement	Global Adjustment; Class A Peak Demand Factor	Consumers; taxpayers for 85% of non-hydro renewables	Two-way (negative months occur)	Monthly estimates; RPP reset May/November	Class A pays on peak contribution and can curtail; Class B households pay the residual per kWh
United Kingdom	S, P, G	CfD levy (ILR/OLR); RO buy-out; price cap	Consumers; Exchequer for 75% of domestic RO costs 2026–29	Two-way (net generator payments in 2021–22)	Quarterly levy rates; cap quarterly	Uniform per-kWh levy; no class differentiation inside CfD
Australia	S, G	LRET certificates; Capacity Investment Scheme collar	Consumers (LRET); taxpayers (CIS)	CIS partial two-way (90%/50%)	Certificate surrender annual	Retail pass-through undifferentiated; large users contract directly
Nordics	S, P	Elcertificates (closed 2021); grid tariffs	Consumers	Certificate price two-way in market, one-way in obligation	Annual quota	Uniform; large users hedge financially

Market	Dominant channels	Named schemes	Payer	Direction	Reset lag	Household vs large-user incidence
South Africa	R, G (proposed)	REIPPPP PPAs in tariff; SAWEM Legacy Charge (draft)	Consumers	Legacy charge two-way by formula	Annual (NERSA)	Uniform legacy charge on all consumers in the draft rules
Mexico	R	CFE Basic Supply subsidy transfers	Taxpayers	One-way (budget)	Budget cycle	89% of users subsidized; incidence set by the tax system
Philippines	S, G	FIT-All charge	Consumers	Two-way formula, suspended collection 2022–24	ERC decisions per filing	Uniform per-kWh on all grid consumers

Table 6: Prevalence by covered volume, 2024. TWh and percent of world electricity consumption (30,856 TWh).

Channel	Covered (TWh)	Share (%)	Range (%)
Spot / merchant (S)	2,765	9.0	6.9–11.4
Private contracts (P)	6,502	21.1	19.0–21.1
Publicly backed contracts (G)	649	2.1	2.0–2.2
Regulated tariffs (R)	10,740	34.8	30.8–39.5
Classified, all channels	20,656	66.9	
Overlap (in two channels before de-overlap)	649	2.1	
Unknown (not classified from published sources)	10,200	33.1	

Notes: Covered volumes are the terawatt-hours published as settled under each channel, summed from `data/prevalence_inputs.csv` (26 entries, 11 primary-verified). Shares use Ember 2024 world demand as the denominator. Ranges vary each input between its low and high entries independently, so the range endpoints do not sum to the central shares. Source: Desk computation (Reconstructed, B).

The EU-level rule sits above the member rows: Regulation (EU) 2024/1747 requires two-way contracts for difference for new publicly supported wind, solar, geothermal, reservoir-less hydro and nuclear investment from 17 July 2027, which will convert the remaining one-way premium designs in Germany, France and Spain into Ontario-form settlements for new capacity.

9 Prevalence: how much of world electricity each channel settles

This section replaces whole-country classification with covered volumes. The inputs are the 26 entries of `data/prevalence_inputs.csv` (11 primary-verified), each a published volume of electricity settled under one channel in one market, entered once under its revenue-determining channel. Table 6 and Figure 4 report the result against Ember's 2024 world demand of 30,856 TWh.

The headline is the ordering. Regulated tariffs are the largest classified channel at 10,740 TWh (34.8 percent of world consumption; range 30.8–39.5). Private contracting follows at 6,502 TWh (21.1 percent), driven by China's 6,200 TWh of banded market trading. Spot and merchant settlement, the channel that dominates policy discussion, covers 2,765 TWh (9.0 percent; range 6.9–11.4), and publicly backed contracts settle 649 TWh (2.1 percent; range 2.0–2.2). Two-thirds of world consumption (20,656 TWh, 66.9 percent) can be classified from published covered volumes; 10,200 TWh (33.1 percent) cannot.

The overlap is exactly the publicly backed volume: all 649 TWh of Channel G energy also trades at a spot or pool price, so any accounting that adds market-traded volume and supported volume counts those terawatt-hours twice. Germany's 238.0 TWh of EEG-entitled generation is the largest single overlap block; the UK's CfD fleet, at 22.2 TWh against UK demand of about 319 TWh, shows how small the settled slice can be inside a system whose policy debate is dominated by the scheme.

Why the covered-volume answer differs from the scoping review's whole-country split (spot-led systems 19 percent, gap-charge systems 72 percent, cost-regulated systems 8 percent) is mechanical, and each part of the difference is visible in the inputs. First, the whole-country method credited a gap charge with a country's entire consumption; the covered method credits it only with the supported volume, which is 46 percent of consumption in Germany, 7 percent in the UK, and unpublished for Spain and Poland. Second, the whole-country method classified the United States, China and India by their most visible market institution; the covered method finds 2,200 TWh of US regulated volume, 3,866

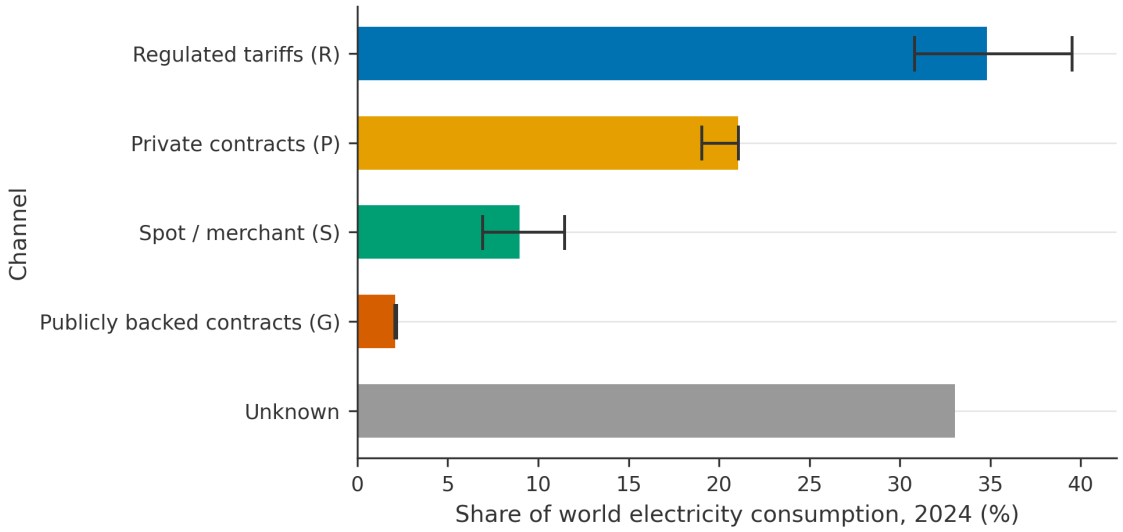


Figure 4: Shares of 2024 world electricity consumption by settlement channel, covered-volume method. Whiskers show the range from varying each input between its low and high entries. Source: (RECONSTRUCTED, B).

TWh of Chinese non-traded volume and 1,592 TWh of Indian long-term PPA volume inside those systems. Third, the whole-country method had no residual: the covered method leaves a third of world consumption unclassified, composed of the rest-of-world block outside the markets examined (5,645 TWh), the EU volume whose merchant/private split is not published (2,376 TWh), Japan (1,027 TWh, whose administrator publishes purchase costs but not purchased volume), and smaller residuals in the UK, India, the United States, the Philippines, Russia, Australia and Mexico.

Sensitivity. Three classification choices move the results materially. (i) China: if China's banded market-traded volume were classified as merchant (S) rather than private contracting (P), S would rise to 8,965 TWh (29.1 percent) and P would fall to 302 TWh (1.0 percent). We classify it as P because trading is anchored to provincial coal benchmarks inside a ±20 percent band and dominated by medium- and long-term contracts; readers who treat any exchange-cleared volume as spot should use the alternative row. This single choice is worth 20.1 percentage points of world demand, larger than the entire classified spot channel. (ii) The United States: the merchant share of non-ERCOT RTO load is assumed at 60 percent (range 40–90), the widest range in the table, and drives most of the S range. (iii) Reference years: all inputs are 2024 except India's (FY2024-25, computed on a generation basis against a demand denominator, which inflates India's residual) and Italy's supported volume (2025). Neither moves any channel share by more than 0.2 points.

10 Discussion

The charge is an automatic stabilizer, and it moves inversely to the price it replaces. In every two-way system examined, the gap charge is a decreasing function of the spot price, mechanically. The worked example's pass-through coefficient of 0.30 means the average consumer keeps 30 cents of each dollar of wholesale price movement, in both directions, with the levy absorbing the rest. The observed systems show the same signature at larger amplitude: Japan's surcharge fell by more than half in the year after the FY2022 price spike and then rose above its pre-crisis level as prices

normalized; the UK's CfD account paid money back to suppliers at the peak of the 2021–22 crisis; Ireland's PSO levy went to minus €491.25M; France's CSPE compensation went negative by €2.7B in a single half-year; Ontario's GA has been negative in 18 months since 2005. A policy debate that treats these charges as a cost of renewables misreads the sign half the time: in high-price states of the world the charge is a consumer rebate financed by contracted generators, and its stabilizing value is largest exactly when retail politics is most strained.

Price signals survive only on the uncovered slice. Blunting is proportional to coverage. Where coverage is near-total (Ontario's GA settles almost the whole fleet; Korea's pool plus regulated tariffs), the marginal signal reaching consumers is weak and arrives through administrative resets. Where coverage is partial (the UK's CfDs at 7 percent of demand), the levy is a small wedge on a bill that still moves with the market. The prevalence results put the world's blunted share in perspective: contracted and regulated channels settle 58 percent of world consumption between them (Channels P, G and R), so the textbook spot signal is the minority experience globally, and efficiency analysis that assumes marginal-cost pricing for world electricity assumes the exception.

Ratepayer or taxpayer is a distributional choice, not an economic one. The same euro of gap is collected from electricity consumers in Ontario, the UK, Japan and the Philippines, and from general taxation in Germany (since 2022), Australia's CIS and Mexico. Levy funding loads the cost onto electricity consumption, which is regressive per kWh but preserves the link between the charge and the system that caused it; budget funding spreads it over the tax base and removes it from the marginal price of electricity, which matters more as electrification policy tries to make electricity cheap relative to gas. Germany's 2022 switch, taken at the peak of the price crisis, and the UK's 2026 decision to move 75 percent of domestic Renewable Obligation costs to the Exchequer both moved incidence without changing the underlying contracts by one clause. Ontario's 2021 shift of 85 percent of non-hydro renewable costs to the tax base is the same instrument at smaller scale.

Deferral is the same charge with a longer lag and interest. Spain's tariff deficit, India's regulatory assets, China's REDF arrears and KEPCO's pool losses are one device in four legal forms: the gap was computed, or computable, and the system chose not to collect it in the period. The stock figures are of Ontario-decade scale (Spain's deficit exceeded €30B by end-2012; six Indian jurisdictions carry about Rs 1.77 lakh crore; China's reported fund deficit is RMB 821B), and each carries financing costs that a current levy would not. India's Supreme Court has now imposed the correction (liquidation timelines from April 2024), as Spain did through securitization a decade earlier. A gap charge that is not collected does not disappear; it becomes public or quasi-public debt with the electricity consumer as the eventual payer, and the worked example's deferral arithmetic ($62.5M carried per quarter per $250M of annual cost) is the small-scale version of the same compounding.

Exemptions decide who the stabilizer stabilizes. Energy-intensive exemptions and caps recur in Germany (BesAR), Italy (the ASOS concessions line, 24.17 points of the charge), Poland (consumption-banded capacity levy) and Japan (large-user reductions, secondary evidence). In the worked example, exempting 40 percent of consumption from the levy base raises the household's levy cost in the decline case from $95 to $158.33, a 67 percent increase, without changing the total by a dollar. Because large users also have the greatest ability to contract around volatility privately, exemptions systematically transfer price risk from the class best able to bear it to the class least able, and they do so invisibly, inside a charge whose total is unchanged.

11 Limitations

Four limits bound what this paper claims. First, coverage of mechanisms is not coverage of countries: the market sections examine the largest systems and one contrasting case per region, and the prevalence estimate's unknown share (33.1 percent of world consumption) is dominated by systems not examined at all. The unknown share is reported, not modeled away. Second, several load-bearing figures are secondary-only, and Appendix A names them: the merchant share of non-ERCOT US load (an assumption with a stated range), India's channel split and regulatory-asset aggregates beyond PFC's state table, Korea's pool statistics, China's fund deficit (a Bruegel figure attributed to Ministry of Finance data), the Philippines' FIT-All rate path, Russia's regulated volume, and South Africa's consumption. The prevalence ranges are built from these grades, but a range around a secondary figure is not verification. Third, channels are assigned by the revenue-determining arrangement; real fleets mix them (a generator with a CfD on part of its output and merchant exposure on the rest is entered once), so channel shares measure settlement of volume, not counts of plants or firms. Fourth, the worked example is an accounting device with flat demand and a single period price; it establishes incidence arithmetic, not welfare effects, and its pass-through coefficient is a property of its assumed channel shares, not an estimate for any real market.

Two falsifiable implications are stated for the record in Section 5: that measured levy changes across the markets in Appendix A move against spot prices with the stated lags, and that the covered volumes in `data/prevalence_inputs.csv` sum, market by market, to no more than the consumption of the markets they describe. Both were constructed to be checkable from the saved data files.

12 Conclusion

When wholesale electricity prices fall, the money does not vanish; it is reassigned by contract. On the evidence assembled here, the reassignment runs mainly through four channels whose world weights are now measurable: regulated tariffs settle about 35 percent of world consumption, private contracts about 21 percent, merchant exposure about 9 percent, and publicly backed gap contracts about 2 percent, with a third of world consumption not classifiable from published volumes. The gap charge, the arrangement that Ontario's Global Adjustment exemplifies, is small as a share of volume and large as a share of bill politics, because it is the one component designed to move against the market price, visibly, on a published schedule. Systems that refused to collect it currently (Spain, India, China, Korea) hold it instead as deficit, regulatory asset, arrears or utility debt. The design questions that matter for the next decade are the ones this paper's five variables isolate: whether the settlement is two-way, how fast it resets, which customer classes are inside the base, and whether the payer is the ratepayer or the taxpayer. The EU's 2027 two-way mandate and South Africa's draft legacy charge suggest the Ontario form, a formula-settled, consumer-funded, two-way gap charge, is becoming the default template for new support. Whether it should be depends on the incidence arithmetic, which is now on the table.

References

Severin Borenstein, James B. Bushnell, and Frank A. Wolak. Measuring market inefficiencies in california's restructured wholesale electricity market. *American Economic Review*, 92(5):1376–1405, 2002.

James B. Bushnell, Erin T. Mansur, and Celeste Saravia. Vertical arrangements, market structure,

and competition: An analysis of restructured us electricity markets. *American Economic Review*, 98(1):237–266, 2008.

María Paz Espinosa. Understanding the electricity tariff deficit: Lessons from spain. Working paper, Universidad del País Vasco, 2013.

Natalia Fabra. The energy transition: An industrial economics perspective. *International Journal of Industrial Organization*, 79:102734, 2021.

Natalia Fabra and Mar Reguant. Pass-through of emissions costs in electricity markets. *American Economic Review*, 104(9):2872–2899, 2014.

Paul L. Joskow. Lessons learned from electricity market liberalization. *The Energy Journal*, 29 (Special Issue 2):9–42, 2008.

Philippe Menanteau, Dominique Finon, and Marie-Laure Lamy. Prices versus quantities: Choosing policies for promoting the development of renewable energy. *Energy Policy*, 31(8):799–812, 2003.

David Newbery. Missing money and missing markets: Reliability, capacity auctions and interconnectors. *Energy Policy*, 120:695–703, 2018.

Jos Sijm, Karsten Neuhoff, and Yihsu Chen. Co2 cost pass-through and windfall profits in the power sector. *Climate Policy*, 6(1):49–72, 2006.

A Data appendix: sources and verification status

Every load-bearing figure in the paper is listed with the source read for this paper and its verification status. **Primary-verified** means the figure or mechanism was read in a document of the regulator, market or system operator, legislator, or scheme administrator named. **Secondary-only** means the best source located was reporting, an aggregator, or another institution's compilation; those figures are flagged at first use in the text and carry ranges in the prevalence inputs. **Gap** means no usable figure was located; the item is named so that the absence is auditable.

Figure or mechanism	Source read	Status
World demand 2024, 30,856 TWh; China 10,066; US 4,401; EU-27 2,727; India 2,054	Ember, Global Electricity Review 2025, ch. 3.1	Primary-verified
Country demand below the top four (Japan 1,027; Brazil 775; Canada 654; Korea 628; Russia 1,178; Mexico 390; Nordics 390; Australia 265)	Ember GER 2025 as reproduced (Set A vintage); two Ember vintages conflict below the top four	Secondary-only
Ontario GA C$94.5B vs commodity C$47.1B, Oct 2016–Sep 2026; GA negative in 18 months over 2005–2026	Estate workbook of IESO settlement data, recomputed (Section 7.8)	Primary-verified
Ontario GA mechanics (Class A PDF, Class B per-kWh, estimate ladder)	IESO market documentation; IESO Market Manual 5.6	Primary-verified
Ontario demand 2024, 140.4 TWh	IESO Reliability Outlook, June 2025	Primary-verified

Figure or mechanism	Source read	Status
85% of non-hydro renewable costs to the tax base from 2021	IESO/Ontario programme documentation	Primary-verified
Two-way CfD mandate for new supported low-carbon from 17 Jul 2027	Regulation (EU) 2024/1747, Article 19d	Primary-verified
Germany EEG volume 238.0 TWh and €18.5B paid, 2024; levy zero from Jul 2022	Bundesnetzagentur EEG statistics and monitoring	Primary-verified
Germany 2017 EEG levy peak 6.88 ct/kWh	Secondary compilations	Secondary-only
France supported volume 2024 (premium 15.7 TWh = 25% of total)	Commission de régulation de l'énergie	Primary-verified
France ARENH end date 31 Dec 2025	CRE	Primary-verified
ARENH volume/price pair (100 TWh at €42)	Secondary	Secondary-only
France CSPE compensation −€2.7B, H1 2023	EDF half-year presentation	Primary-verified
Spain tariff deficit stock (over €30B by end-2012)	Espinosa (2013) and CNE-derived secondary	Secondary-only
Italy ASOS shares and system charges, Q3 2025; support €9.3B on 50.2 TWh	ARERA quarterly determination; GSE	Primary-verified
Poland RES charge PLN 7.3/MWh (2026), capacity levy, household tariff	URE 2026 determinations	Primary-verified
Poland price freeze funding route	Secondary	Secondary-only
UK CfD net payments Q4 2021 (£133.7M) and supplier payback (£39.2M); ILR £0	LCCC scheme data	Primary-verified
UK CfD generation 22.2 TWh 2023/24	LCCC	Primary-verified
UK final demand 2024, 318.7 TWh	DESNZ, Digest of UK Energy Statistics	Primary-verified
UK RO cost shift: 75% of domestic RO costs, 3 years from Apr 2026, £7B	DESNZ decision	Primary-verified
Ireland PSO levy −€491.25M (2022/23) vs +€263.7M (2021/22)	CRU Decision CRU202277	Primary-verified
Nordic certificate scheme: closure 2021, 28.4 TWh target met 2019	Norwegian government/NVE	Primary-verified
US retail sales 3,975 TWh; energy-only suppliers 14.9%	EIA Electricity Profile 2024	Primary-verified
Two-thirds of US load in RTO regions	FERC	Primary-verified
ERCOT: 90% of Texas load; cap $5,000/MWh from 2022	PUCT/ERCOT rule documents	Primary-verified
ERCOT 2024 energy, 482 TWh	Secondary compilation of ERCOT data	Secondary-only
Merchant share of non-ERCOT RTO load (60%, range 40–90)	Assumption, stated in Section 9	Assumed
Fuel adjustment clause mechanics	Kentucky PSC rule 807 KAR 5:056	Primary-verified

Figure or mechanism	Source read	Status
NY CES structure; ZEC cap 27,618 GWh/yr; OREC awards 810 + 924 MW	NY PSC Case 15-E-0302; NYSERDA	Primary-verified
NY CES per-kWh surcharge rates	Not published in sources read	Gap
NJ OREC surcharge formula	NJ Administrative Code §14:8-6.6	Primary-verified
NJ collected OREC surcharge rate	No project operating	Gap
China consumption 9.85 trillion kWh; market-traded 6.2 trillion kWh (63%), 2024	National Energy Administration	Primary-verified
China ±20% band; residential price stability	NDRC/NEA 2021 reform documents	Primary-verified
China renewable surcharge RMB 0.019/kWh, frozen since 2016	Secondary	Secondary-only
China REDF deficit RMB 821B (2012–2024)	Bruegel WP 15/2026, attributed to MOF 2025; MOF document not read	Secondary-only
China Document 136 mechanism price (from Jun 2025)	Secondary reporting of NDRC/NEA text; Chinese primary not read	Secondary-only
China FIT-paid volume, TWh	Not published in sources read	Gap
India ACS–ARR gap Rs 0.19/kWh; subsidy 20.21%; borrowings Rs 7,52,677 crore	PFC, Report on Performance of Power Utilities	Primary-verified
India regulatory-asset stocks by state (Delhi, TN, Rajasthan, others)	PFC state tables	Primary-verified
India all-India regulatory-asset total	Supreme Court judgment figure Rs 1.74 lakh crore; ICRA estimate Rs 3 lakh crore	Secondary-only
India generation 1,829.69 TWh; exchange volume 143.75 BU (7.9%)	CERC data via secondary reporting	Secondary-only
India RPO/RCO obligation 29.91% FY2024-25	Ministry of Power	Primary-verified
Japan surcharge FY2024 ¥3.49, FY2025 ¥3.98; components and sales denominators	METI press releases, Mar 2024 and Mar 2025	Primary-verified
Japan surcharge FY2023 ¥1.40	Renewable Energy Institute reporting of METI determination	Secondary-only
Japan FIT/FIP purchased volume, TWh	METI English releases publish costs, not volume	Gap
Korea RPS obligation 13.5/14/15%, 25% by 2030	Korea Energy Agency	Primary-verified
Korea SMP averages; RPS volume 63.8 TWh; REC price; Oct 2024 industrial +9.7%	Secondary compilations	Secondary-only
Brazil CDE 2025: R$49.227B total; R$46.886B consumer-paid	ANEEL Resolução Homologatória 3.484/2025	Primary-verified
Brazil ACL 39% / ACR 61% of consumption, 2024	CCEE Balanço 2024	Primary-verified
Brazil tariff flag mechanics	Secondary	Secondary-only
Australia CIS collar (90% floor top-up, 50% ceiling clawback, 15 years)	DCCEEW CIS Tender 8 guidelines	Primary-verified
Australia CIS size 40 GW (uplifted Jul 2025)	DCCEEW	Primary-verified

Figure or mechanism	Source read	Status
Australia LRET target 33,000 GWh; RPP 17.91% (2025)	Clean Energy Regulator	Primary-verified
Australia NEM consumption (178 TWh operational vs 213 TWh total)	AEMO; AER (different quantities, both published)	Primary-verified
South Africa MYPD6 generation allowable revenues	NERSA Reasons for Decision	Primary-verified
South Africa SAWEM Legacy Charge	NERSA pricing methodology consultation; NTCSA draft CPA Conduct Rules (both pre-final)	Primary-verified (draft status)
Mexico: State $\geq$54% of energy; CFE subsidy transfer framework	DOF Energy Sector Programme 2025-2030; SHCP transfer rules	Primary-verified
Mexico 2025 subsidy MXN 84,805M; 89% of users; 47% of bill	CFE director-general testimony as reported	Secondary-only
Philippines generation 2024, 126,941 GWh	Philippines DOE power statistics	Primary-verified
Philippines FIT-All rates 2022–2025 and collection suspension	ERC decisions as reported; decision texts not read	Secondary-only
Philippines FIT fleet 84 plants, 1,707.63 MW	Secondary	Secondary-only
Russia regulated volume share	Not separately verified in this pass	Secondary-only
South Africa consumption 210 TWh	Not confirmed in Ember sources read	Secondary-only
Rest of Canada: cost-of-service systems; no GA-style charge elsewhere	Provincial synthesis; absence to the depth searched	Secondary-only